

The empirical recurrence for OEIS A295037: recovering a conjecture that is not written down

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Abstract

OEIS A295037 records “Empirical recurrence of order 91”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 92$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 91, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 91 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295037 is “Number of $n \times 5$ 0..1 arrays with each 1 horizontally or vertically adjacent to 1 or 2 1s.”. It has offset 1 and begins

12, 280, 3853, 59126, 898493, 13706771, 209181266, 3184995895, ...

The entry states:

Empirical recurrence of order 91 (see link above)

As of the “Last modified” line on the live entry (Nov 12 2017 20:03:33, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 92$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 92$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 184$ exact terms of the model returns order 91 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{91} c_i a(n-i),$$

c_1	10	c_{24}	270927688719	c_{47}	−2812148003689	c_{70}	1041337901
c_2	18	c_{25}	−408244824494	c_{48}	1808160148221	c_{71}	−536317014
c_3	736	c_{26}	293758183202	c_{49}	163963577890	c_{72}	160677714
c_4	2250	c_{27}	1045513924080	c_{50}	156886286738	c_{73}	−125018577
c_5	13808	c_{28}	−3073604382945	c_{51}	1144913029242	c_{74}	59038725
c_6	7108	c_{29}	3351801666376	c_{52}	−1056189301699	c_{75}	−21103726
c_7	−25321	c_{30}	−5363694584542	c_{53}	1473063088923	c_{76}	10745752
c_8	−1150234	c_{31}	2620506549365	c_{54}	−291037218415	c_{77}	−5309703
c_9	−3397594	c_{32}	−395079131831	c_{55}	−132881871818	c_{78}	1462125
c_{10}	15432948	c_{33}	817576922389	c_{56}	223363971990	c_{79}	−390047
c_{11}	43775711	c_{34}	5306966578996	c_{57}	171892923733	c_{80}	111895
c_{12}	−74842407	c_{35}	487263259850	c_{58}	45938951141	c_{81}	2350
c_{13}	−288949088	c_{36}	3341503713848	c_{59}	66247931426	c_{82}	−2987
c_{14}	−80592544	c_{37}	1015590266461	c_{60}	−85632139323	c_{83}	4935
c_{15}	1447977146	c_{38}	−4101973676071	c_{61}	16322212873	c_{84}	−1766
c_{16}	2448447349	c_{39}	2110036461980	c_{62}	68129658592	c_{85}	58
c_{17}	−5283816347	c_{40}	−5111872302186	c_{63}	−55404973858	c_{86}	−101
c_{18}	−11237267334	c_{41}	4583436499699	c_{64}	16399510183	c_{87}	37
c_{19}	9563696210	c_{42}	1658809035981	c_{65}	322885188	c_{88}	−11
c_{20}	33531362055	c_{43}	853749271023	c_{66}	−2078208327	c_{89}	2
c_{21}	9761278145	c_{44}	5440401099915	c_{67}	−3820350739	c_{90}	−1
c_{22}	−140306423700	c_{45}	−5977573922621	c_{68}	2310101434	c_{91}	1
c_{23}	10658769122	c_{46}	3864636148197	c_{69}	−1265795505		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 91, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $91 + 4$ terms removed returns order 91 as well, so $\deg q_{\min} = 91 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $91 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 91 that this sequence satisfies — and that family turns out to have exactly one member.

References

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